# 5-root-cause-analysis-uber

November 25, 2024

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[2]:
```
df = pd.read_csv('uber-data.csv', parse_dates=[4, 5])
df.head()
```

[2]:
```
   Request id Pickup point  Driver id          Status   Request timestamp  \
0         619      Airport        1.0  Trip Completed 2016-11-07 11:51:00
1         867      Airport        1.0  Trip Completed 2016-11-07 17:57:00
2        1807         City        1.0  Trip Completed 2016-12-07 09:17:00
3        2532      Airport        1.0  Trip Completed 2016-12-07 21:08:00
4        3112         City        1.0  Trip Completed 2016-07-13 08:33:16

       Drop timestamp
0 2016-11-07 13:00:00
1 2016-11-07 18:47:00
2 2016-12-07 09:58:00
3 2016-12-07 22:03:00
4 2016-07-13 09:25:47
```

[3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6745 entries, 0 to 6744
Data columns (total 6 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Request id         6745 non-null   int64
 1   Pickup point       6745 non-null   object
 2   Driver id          4095 non-null   float64
 3   Status             6745 non-null   object
 4   Request timestamp  6745 non-null   datetime64[ns]
 5   Drop timestamp     2831 non-null   datetime64[ns]
dtypes: datetime64[ns](2), float64(1), int64(1), object(2)
memory usage: 316.3+ KB
```

[4]:
```
df.isnull().sum()/len(df)*100
```

[4]:
```
Request id            0.000000
Pickup point          0.000000
Driver id            39.288362
Status                0.000000
Request timestamp     0.000000
Drop timestamp       58.028169
dtype: float64
```

[5]:
```
df['RequestHour'] = df['Request timestamp'].dt.hour
df.head()
```

[5]:
```
   Request id Pickup point  Driver id          Status   Request timestamp  \
0         619      Airport        1.0  Trip Completed 2016-11-07 11:51:00
1         867      Airport        1.0  Trip Completed 2016-11-07 17:57:00
2        1807         City        1.0  Trip Completed 2016-12-07 09:17:00
3        2532      Airport        1.0  Trip Completed 2016-12-07 21:08:00
4        3112         City        1.0  Trip Completed 2016-07-13 08:33:16

       Drop timestamp  RequestHour
0 2016-11-07 13:00:00           11
1 2016-11-07 18:47:00           17
2 2016-12-07 09:58:00            9
3 2016-12-07 22:03:00           21
4 2016-07-13 09:25:47            8
```

[6]:
```
times = [0, 4, 9, 16, 21, 24]
values = ['Dawn', 'Early Morining', 'Noon', 'Late Evening', 'Night']
df['TimeSlot'] = pd.cut(x=df['RequestHour'], bins=times, labels=values)
df
```

[6]:
```
      Request id Pickup point  Driver id             Status  \
0            619      Airport        1.0     Trip Completed
1            867      Airport        1.0     Trip Completed
2           1807         City        1.0     Trip Completed
3           2532      Airport        1.0     Trip Completed
4           3112         City        1.0     Trip Completed
…            …            …          …                  …
6740        6745         City        NaN  No Cars Available
6741        6752      Airport        NaN  No Cars Available
6742        6751         City        NaN  No Cars Available
6743        6754         City        NaN  No Cars Available
6744        6753      Airport        NaN  No Cars Available

      Request timestamp      Drop timestamp  RequestHour        TimeSlot
0   2016-11-07 11:51:00 2016-11-07 13:00:00           11            Noon
```

```
1    2016-11-07 17:57:00 2016-11-07 18:47:00           17    Late Evening
2    2016-12-07 09:17:00 2016-12-07 09:58:00            9  Early Morining
3    2016-12-07 21:08:00 2016-12-07 22:03:00           21    Late Evening
4    2016-07-13 08:33:16 2016-07-13 09:25:47            8  Early Morining
…                    …                   …          …              …
6740 2016-07-15 23:49:03                 NaT          23          Night
6741 2016-07-15 23:50:05                 NaT          23          Night
6742 2016-07-15 23:52:06                 NaT          23          Night
6743 2016-07-15 23:54:39                 NaT          23          Night
6744 2016-07-15 23:55:03                 NaT          23          Night

[6745 rows x 8 columns]
```

[7]:
```
df['Status'].value_counts()
```

[7]:
```
Trip Completed       2831
No Cars Available    2650
Cancelled            1264
Name: Status, dtype: int64
```

[8]:
```
df['Cab Availability'] = df['Status'].apply(
    lambda x: 'Available' if x=='Trip Completed' else 'Not Available')
df
```

[8]:
```
      Request id Pickup point  Driver id             Status  \
0            619      Airport        1.0     Trip Completed
1            867      Airport        1.0     Trip Completed
2           1807         City        1.0     Trip Completed
3           2532      Airport        1.0     Trip Completed
4           3112         City        1.0     Trip Completed
…           …            …           …                …
6740        6745         City        NaN  No Cars Available
6741        6752      Airport        NaN  No Cars Available
6742        6751         City        NaN  No Cars Available
6743        6754         City        NaN  No Cars Available
6744        6753      Airport        NaN  No Cars Available

       Request timestamp      Drop timestamp  RequestHour        TimeSlot  \
0    2016-11-07 11:51:00 2016-11-07 13:00:00           11            Noon
1    2016-11-07 17:57:00 2016-11-07 18:47:00           17    Late Evening
2    2016-12-07 09:17:00 2016-12-07 09:58:00            9  Early Morining
3    2016-12-07 21:08:00 2016-12-07 22:03:00           21    Late Evening
4    2016-07-13 08:33:16 2016-07-13 09:25:47            8  Early Morining
…                    …                   …          …              …
6740 2016-07-15 23:49:03                 NaT          23          Night
6741 2016-07-15 23:50:05                 NaT          23          Night
6742 2016-07-15 23:52:06                 NaT          23          Night
```

```
6743 2016-07-15 23:54:39                NaT          23          Night
6744 2016-07-15 23:55:03                NaT          23          Night

     Cab Availability
0           Available
1           Available
2           Available
3           Available
4           Available
…                 …
6740    Not Available
6741    Not Available
6742    Not Available
6743    Not Available
6744    Not Available

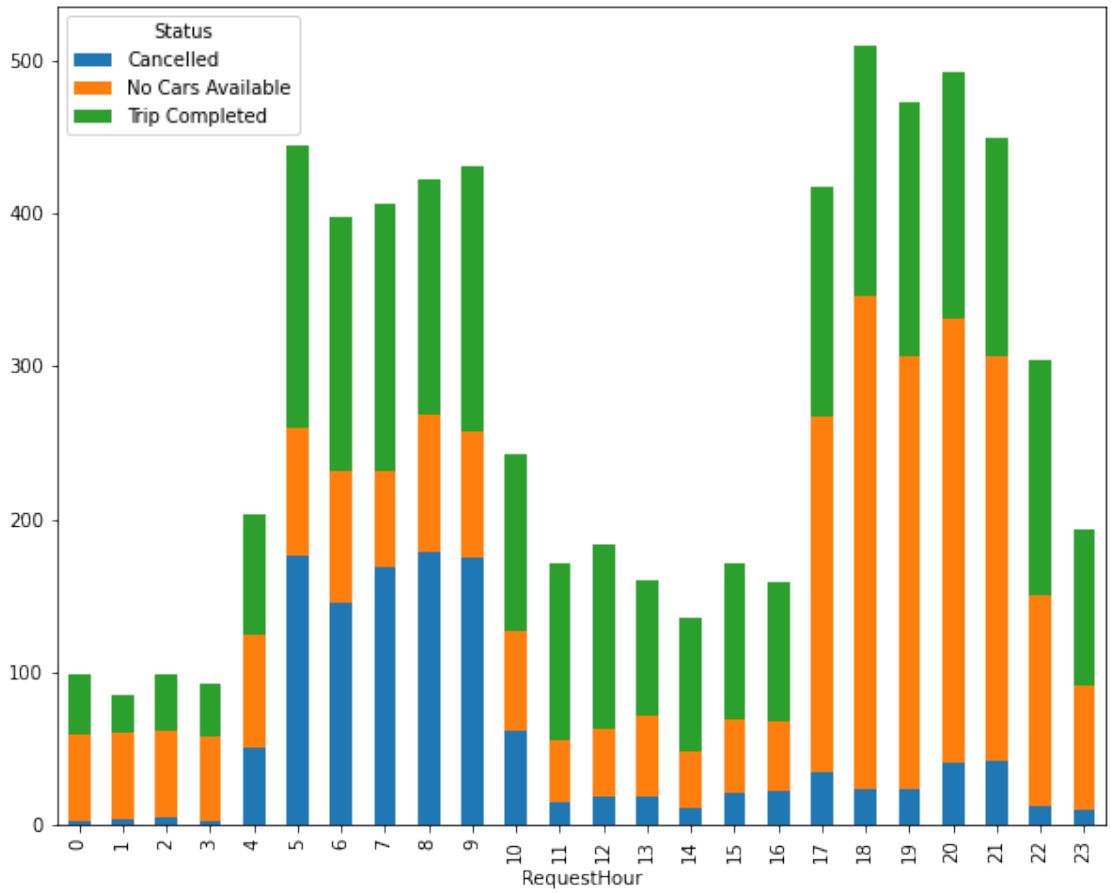
[6745 rows x 9 columns]
```

[11]:
```
df['Cab Availability'].value_counts(normalize=True)*100
```

[11]:
```
Not Available    58.028169
Available        41.971831
Name: Cab Availability, dtype: float64
```

[14]:
```
# What is the frequency of requests that get cancelled or show no cars␣
 ↪available in each hour?

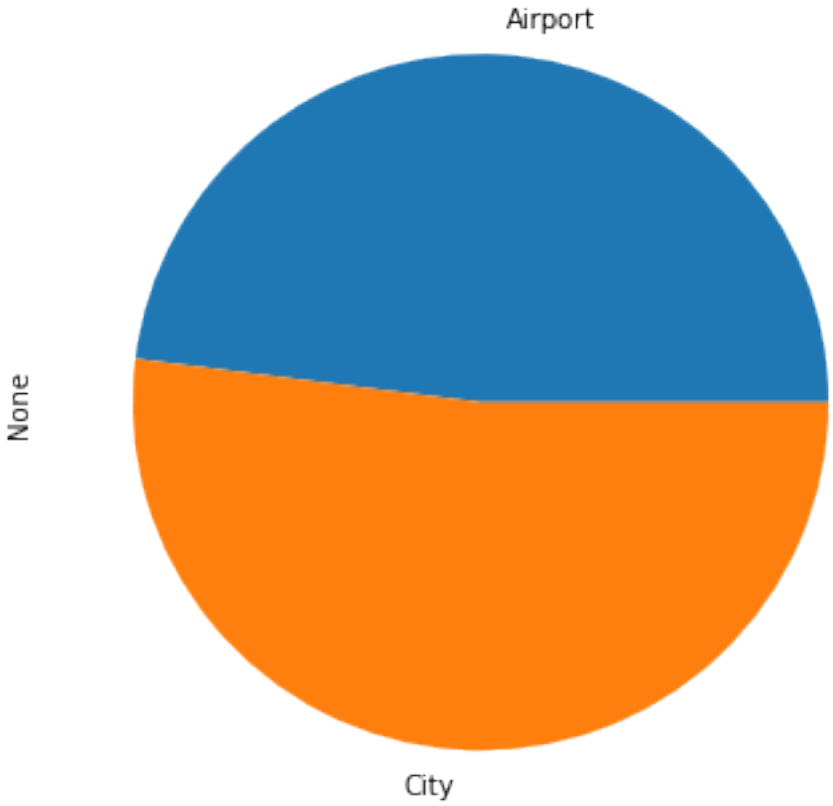
df.groupby(['RequestHour', 'Status']).size().unstack().plot(kind='bar',␣
 ↪stacked=True, figsize=(10, 8))
plt.show()
```




**Observation**

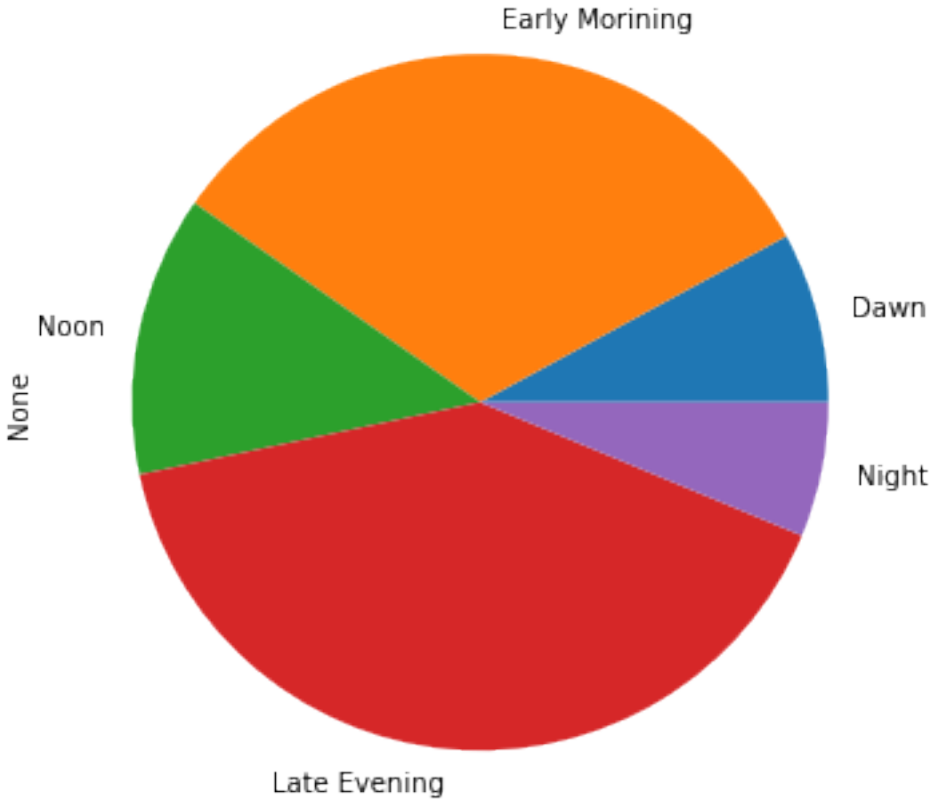
- From 5 AM TO 9AM more number of cabs got cancelled
- From 5PM to 9PM No cars are available

[16]:
```
df.groupby(['Pickup point']).size().plot(kind='pie', stacked=True, figsize=(6,␣
↪6), table=True)
plt.show()
```




| | Airport | City |
|---|---|---|
| None | 3238 | 3507 |

[17]:
```
df[(df['Cab Availability'] == 'Not Available')].groupby(['TimeSlot']).size().
↪plot(kind='pie',
                                                                              ␣
↪  figsize=(6, 6), table=True)
plt.show()
```



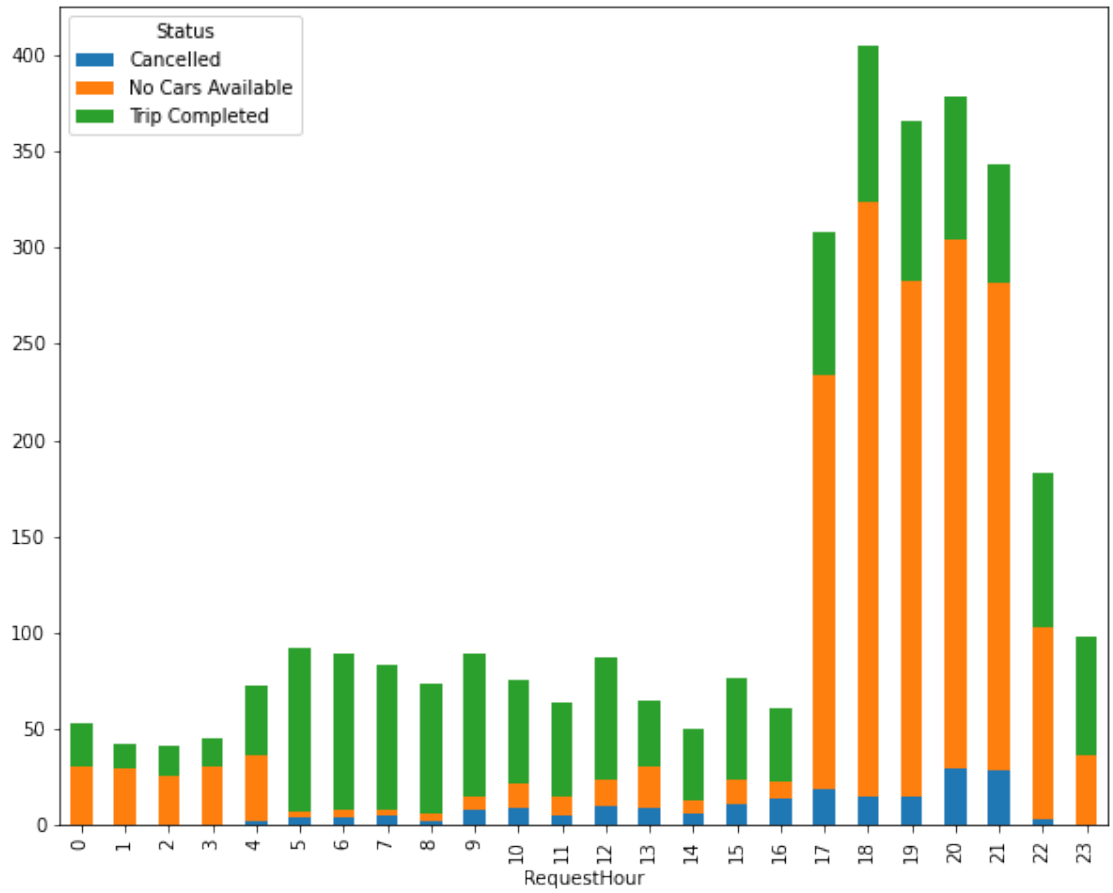

| | Dawn | Early Morining | Noon | Late Evening | Night |
|---|---|---|---|---|---|
| None | 305 | 1249 | 502 | 1558 | 241 |

Observation:

- Late evenings and Early mornings are not recommended for airport -city transport or viceversa

[18]:
```
# Demand supply gap from airport to city

df[(df['Pickup point'] == 'Airport')].groupby(['RequestHour', 'Status']).size().
 ↪unstack().plot(
    kind='bar', stacked=True, figsize=(10, 8))
plt.show()
```

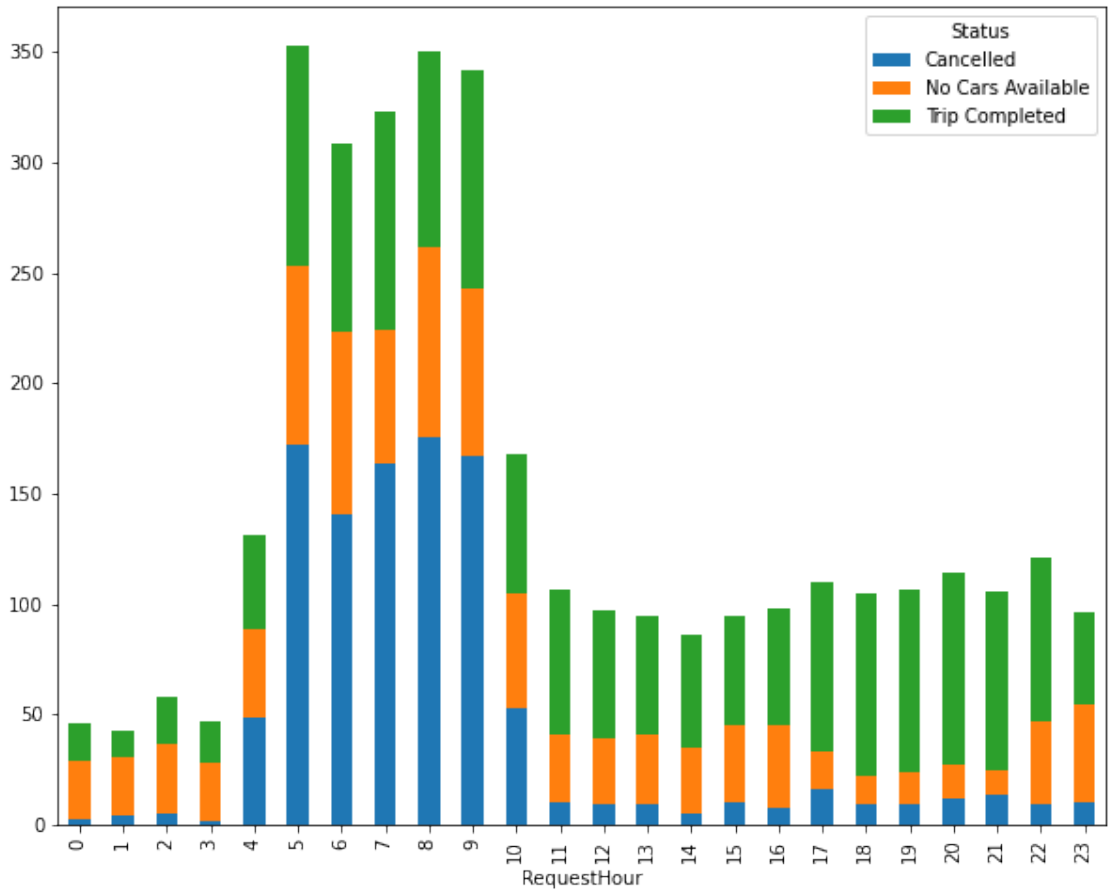



Observation

1. There is very high demand for cabs from airport to city between 5 pm to 9pm but supply is very less due to no cabs avaiable

[19]:
```
# Supply demand gap from city to airport

df[(df['Pickup point'] == 'City')].groupby(['RequestHour', 'Status']).size().
 ↪unstack().plot(
    kind='bar', stacked=True, figsize=(10, 8))
plt.show()
```

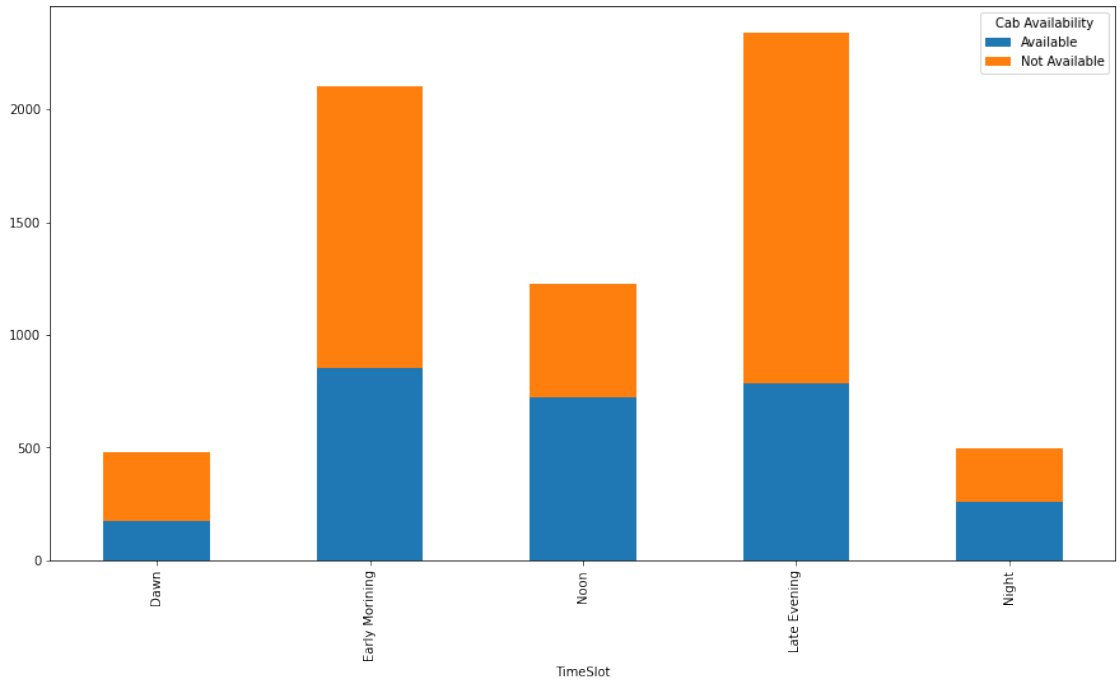



Observation:

- There is very high demand of cabs from city to airport between 5AM to 9AM, But supply is less primarly due to ride cancellations

[20]:
```
# Time slots where the highest gap exists

df.groupby(['TimeSlot', 'Cab Availability']).size().unstack().plot(kind='bar',␣
↪stacked=True,
                                                                  figsize=(15,␣
↪8))
plt.show()
```


Cab Availability
Available
Not Available
2000
1500
1000
500
0
Dawn
Early Morining
Noon
Late Evening
Night
TimeSlot